

NARITA INTL, Japan

WMO: **476860**

Lat: **35.765N** Lon: **140.386E** Elev: **141** StdP: **14.62** Time Zone: **9.00 (E09)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	24.9	27.0	8.5	8.5	39.4	11.8	10.1	39.1	25.7	44.9	22.8	44.1	3.7	330	0.461	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	8	13.3	91.3	78.7	89.3	78.0	86.4	77.2	80.1	87.3	79.0	85.8	78.1	84.5	10.3	220	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	78.6	149.1	85.2	77.1	142.0	82.9	75.6	134.9	81.3	44.0	87.9	42.7	85.8	41.6	84.4	85.6	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	22.6	19.4	17.0	DB	20.3	95.1	2.1	2.0	18.8	96.5	17.6	97.7	16.5	98.8	15.0	100.3	(4)
(5)				WB	18.6	82.8	1.9	1.3	17.3	83.7	16.2	84.5	15.1	85.2	13.7	86.2	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	58.7	38.8	40.4	46.4	55.5	63.6	69.5	77.0	79.2	73.2	63.5	52.8	43.5	(6)
(7)			DBStd	14.74	4.25	5.09	6.02	6.06	4.84	4.65	5.00	4.18	5.52	5.23	5.60	5.26	(7)
(8)			HDD50	1037	347	274	148	20	0	0	0	0	0	0	34	214	(8)
(9)			HDD65	3589	811	689	577	289	83	12	0	0	5	90	365	668	(9)
(10)			CDD50	4223	1	5	36	185	421	585	837	907	696	420	119	11	(10)
(11)			CDD65	1300	0	0	0	4	39	147	372	442	250	45	1	0	(11)
(12)			CDH74	9431	0	0	0	23	206	662	2996	3888	1500	155	1	0	(12)
(13)			CDH80	3019	0	0	0	2	18	120	1050	1416	393	20	0	0	(13)
(14)	Wind		WSAvg	8.0	7.6	8.2	9.0	9.2	8.4	7.6	7.8	7.7	8.2	8.0	7.0	7.0	(14)
(15)	Precipitation		PrecAvg	57.2	2.6	2.6	4.2	4.5	5.1	5.9	5.2	4.8	7.9	8.0	4.1	2.4	(15)
(16)			PrecMax	77.2	5.8	5.0	7.6	8.1	8.4	11.1	11.0	11.3	15.7	23.6	9.8	7.6	(16)
(17)			PrecMin	41.0	0.0	0.4	1.7	1.7	2.5	1.7	0.6	0.6	1.4	1.0	0.3	0.1	(17)
(18)			PrecStd	8.2	1.7	1.3	1.5	1.8	1.6	2.0	3.1	3.1	3.9	5.3	2.2	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		59.3	64.8	70.1	77.1	82.4	87.5	93.3	94.7	90.0	82.5	71.8	66.1	(19)
(20)			MCWB		51.1	56.1	56.2	62.5	66.9	75.1	78.5	79.8	76.5	71.6	62.4	59.3	(20)
(21)		2%	DB		55.2	58.7	65.9	73.0	78.9	82.8	91.2	91.6	87.6	78.4	68.2	60.6	(21)
(22)			MCWB		47.2	51.7	55.1	60.7	66.3	72.1	78.4	79.2	76.7	69.8	60.6	52.7	(22)
(23)		5%	DB		51.6	53.8	62.3	69.8	75.6	80.5	88.2	89.7	84.4	75.0	65.9	57.0	(23)
(24)			MCWB		43.1	45.0	54.2	60.1	65.2	71.0	78.0	78.4	75.7	67.4	59.4	49.4	(24)
(25)		10%	DB		48.4	50.3	58.7	66.5	73.4	77.3	86.1	87.7	82.0	71.9	62.8	53.7	(25)
(26)			MCWB		40.3	42.6	51.0	57.3	63.9	70.1	77.2	78.3	74.5	65.3	56.8	46.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		54.0	58.1	60.7	68.0	72.4	76.7	81.7	82.5	79.4	74.4	66.3	61.5	(27)
(28)			MCDB		57.7	63.3	64.3	71.0	77.4	84.4	89.8	90.8	85.2	79.0	69.5	65.2	(28)
(29)		2%	WB		48.3	52.3	57.7	63.8	69.5	74.6	79.8	80.8	78.2	71.6	62.9	54.5	(29)
(30)			MCDB		52.7	57.2	63.3	69.6	75.4	80.1	87.3	88.2	83.8	76.2	66.8	58.6	(30)
(31)		5%	WB		44.4	47.4	55.2	61.3	67.5	73.3	78.7	79.5	77.0	69.3	60.3	50.7	(31)
(32)			MCDB		49.1	51.9	61.0	67.0	72.9	78.1	86.0	86.3	82.5	73.2	63.9	54.8	(32)
(33)		10%	WB		41.7	44.0	51.6	59.3	65.6	71.6	77.6	78.7	75.8	67.1	58.3	48.0	(33)
(34)			MCDB		46.6	48.4	57.1	64.8	71.1	76.2	84.3	85.0	80.6	70.9	61.8	52.3	(34)
(35)	Mean Daily Temperature Range		MDBR		18.6	17.7	17.8	17.4	15.8	12.9	13.1	13.3	12.4	13.6	16.7	18.3	(35)
(36)		5% DB	MCDBR		22.7	23.1	22.8	21.6	20.5	17.1	17.3	17.1	15.3	16.2	19.2	21.3	(36)
(37)			MCWBR		15.7	16.2	15.4	12.8	10.9	8.4	7.6	7.1	7.1	10.0	13.2	14.9	(37)
(38)		5% WB	MCDBR		19.3	21.1	20.5	17.7	16.9	14.1	15.9	15.3	13.8	13.8	16.1	18.5	(38)
(39)			MCWBR		15.0	16.6	16.3	12.4	11.2	8.2	7.7	7.2	7.1	9.7	12.7	14.7	(39)
(40)	Clear-Sky Solar Irradiance		taub		0.379	0.430	0.498	0.547	0.562	0.589	0.583	0.564	0.502	0.476	0.426	0.383	(40)
(41)			taud		2.242	2.094	1.931	1.861	1.881	1.882	1.945	1.986	2.126	2.120	2.174	2.250	(41)
(42)			Ebn at Noon		252	251	243	239	238	231	231	232	240	233	232	242	(42)
(43)			Edh at Noon		34	44	56	64	63	63	59	56	46	43	36	32	(43)
(44)	All-Sky Solar Radiation		RadAvg		852	1000	1265	1483	1603	1438	1575	1589	1220	941	771	744	(44)
(45)			RadStd		58	105	81	146	165	128	261	151	103	92	70	57	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page